Elements of AIML
Unit 03 – Lecture 01 Notes
Introduction to Machine Learning & Datasets

Tofik Ali

February 28, 2026

Contents

1 Lecture Overview **2**

2 How to use these notes (self-study) **2**

3 Learning Outcomes **2**

4 Keywords **2**

5 Abbreviations and symbols **2**

6 What is machine learning? **3**
6.1 A standard definition (Mitchell) . . . 3
6.2 Algorithm vs model (important distinction) . . . 3
6.3 Why not just write rules? . . . 3

7 ML task types **4**
7.1 Supervised learning . . . 4
7.2 Unsupervised learning . . . 4
7.3 Other task families (brief preview) . . . 4
7.4 Reinforcement learning (preview) . . . 4

8 Dataset basics **4**
8.1 Tabular representation . . . 4
8.2 What does "$X \in \mathbb{R}^{n \times d}$" really mean? . . . 4
8.3 Data types . . . 5
8.4 Dataset quality checklist (what can go wrong?) . . . 5

9 Feature sets and representation **5**
9.1 Why representation matters . . . 5
9.2 Common feature transformations . . . 5
9.3 Preprocessing mindset (ties to labs) . . . 5

10 Worked example: house price prediction **6**
10.1 Worked example: spam classification (conceptual) . . . 6

11 A minimal Pandas inspection snippet (illustrative) **6**

12 Common pitfalls (and how to avoid them) **6**

13 Practice (with answers) **6**

14 Exit questions (with answers) **7**

15 Summary **7**

1 Lecture Overview

This lecture introduces **machine learning** and the basic dataset concepts needed for the rest of the course and the labs. We focus on: tasks, datasets, features/labels, and common data quality issues.

Sources: Mueller and Massaron, *Machine Learning for Dummies*, For Dummies, 2016.;Mitchell, *Machine Learning*, selected chapters on learning basics and evaluation.;Scikit-learn documentation, accessed 2026-02-28.

2 How to use these notes (self-study)

When you study ML, always keep the full pipeline in mind: problem definition → dataset → features/labels → training → evaluation. Most beginner mistakes happen because one of these steps is skipped or assumed. Try the practice questions after each major section.

3 Learning Outcomes

- Define ML and list key task categories.
- Explain datasets in terms of rows/examples and columns/variables.
- Distinguish features from labels and identify common feature data types.
- Recognize common dataset issues and why they matter.

4 Keywords

- **Example/instance:** one row (one data point).
- **Feature:** input variable used for prediction (column in X).
- **Label/target:** output variable to predict (vector y).
- **Supervised learning:** learn from labeled examples.
- **Unsupervised learning:** find structure without labels.
- **Generalization:** performance on unseen but similar data.

5 Abbreviations and symbols

- **ML**: machine learning.
- **EDA**: exploratory data analysis.
- X: feature matrix (inputs), usually written as $X \in \mathbb{R}^{n \times d}$ after encoding.

- y: target/label vector (outputs), e.g., $y \in \mathbb{R}^n$ (regression) or $y \in \{1, \ldots, K\}^n$ (classification).
- n: number of examples (rows), d: number of features (columns), K: number of classes.
- $\hat{y}$: prediction produced by a model, $f(\cdot)$.

6 What is machine learning?

ML builds models from data. A common view is:

- you provide examples,
- you choose a learning method (algorithm),
- you train a model to minimize some error,
- you evaluate on unseen data.

6.1 A standard definition (Mitchell)

A classic textbook definition:

> A computer program is said to learn from **experience** E with respect to some **task** T and **performance measure** P if its performance at task T, as measured by P, improves with experience E.

Example:

- T: classify emails as spam/ham
- E: labeled emails from the past
- P: accuracy/F1 on new (unseen) emails

6.2 Algorithm vs model (important distinction)

- A **learning algorithm** is the procedure we run (e.g., "fit linear regression").
- A **model** is what we get after training (e.g., the learned weights w).

During **training**, the model parameters are estimated from data. During **inference/prediction**, the model is used to produce $\hat{y}$ for a new input x.

6.3 Why not just write rules?

Rules are often:

- too complex to specify (vision, language),
- brittle when the world changes,
- incomplete in edge cases.

ML can learn patterns directly from data, but it introduces new responsibilities:

- collecting and cleaning datasets,
- choosing appropriate evaluation (so we do not fool ourselves),

- monitoring models after deployment (data can change over time).

7 ML task types

7.1 Supervised learning

We learn from labeled examples (x_i, y_i).

- **Regression:** y is numeric (price, temperature).
- **Classification:** y is a class label (spam/ham, disease/no disease).

Typical outputs. In regression, $y \in \mathbb{R}$. In classification, $y \in \{1, \ldots, K\}$ for K classes (binary classification is the case $K = 2$).

7.2 Unsupervised learning

We use only x_i (no labels).

- **Clustering:** group similar examples.
- **Dimensionality reduction:** compress/represent data in fewer variables.

7.3 Other task families (brief preview)

- **Semi-supervised learning:** a small labeled set + a larger unlabeled set.
- **Anomaly detection:** find rare/abnormal points (fraud, faults).
- **Ranking/recommendation:** order items for a user (feeds, products).

7.4 Reinforcement learning (preview)

An agent interacts with an environment and learns from rewards. We cover an overview later in Unit IV.

8 Dataset basics

8.1 Tabular representation

Many problems can be represented as a table:

- rows = examples/instances,
- columns = features + (optional) label column.

We typically denote:

$$X \in \mathbb{R}^{n \times d}, \quad y \in \mathbb{R}^n \text{ or } y \in \{1, \ldots, K\}^n$$

where n is the number of examples and d is the number of features.

8.2 What does "$X \in \mathbb{R}^{n \times d}$" really mean?

Even if your raw dataset has text/categorical columns, most ML models require numeric inputs. So we perform **encoding** (e.g., one-hot for categories, TF-IDF for text) to convert raw values into numerical feature vectors. After encoding, we can treat X as a real-valued matrix.

8.3 Data types

Features can be:

- **numerical** (int/float),
- **categorical** (city, category),
- **ordinal** (ordered categories),
- **text/images/time series** (need a feature representation).

8.4 Dataset quality checklist (what can go wrong?)

Common issues that strongly affect ML results:

- **Missing values:** empty/NaN cells; need imputation or removal.
- **Outliers:** extreme values; may be errors or rare but real cases.
- **Duplicates:** repeated rows can bias training and evaluation.
- **Label noise:** incorrect target values (wrong ground truth).
- **Class imbalance:** one class dominates, making accuracy misleading.
- **Data leakage:** information that would not exist at prediction time gets included.

Most lab experiments (EDA, missing values, outliers, scaling, imbalance) are designed to give hands-on practice with these.

9 Feature sets and representation

9.1 Why representation matters

Two models can differ more due to **features** than due to the choice of algorithm. For example, a "location" feature may need **one-hot encoding** before it can be used by many models.

9.2 Common feature transformations

- **One-hot encoding**: convert a categorical value into a 0/1 vector.
- **Scaling/standardization**: make numerical features comparable (mean 0, std 1).
- **Log transforms**: handle skewed distributions (income, counts).
- **Binning**: convert continuous features into buckets (sometimes useful for interpretation).

9.3 Preprocessing mindset (ties to labs)

In the labs you will practice:

- importing/exporting data (Pandas),
- basic summaries and EDA,
- missing values and outlier handling,
- scaling and encoding.

10 Worked example: house price prediction

Suppose we want to predict house price:

- target y = price,
- features X = area, rooms, location, age.

If "location" is categorical, we cannot directly treat it as a number (that would add fake ordering). We encode it using one-hot encoding (lab experiment on encoding).

10.1 Worked example: spam classification (conceptual)

- Task: classify email as {spam, ham}.
- Raw input: text; label: spam/ham.
- Features: bag-of-words counts, TF-IDF vectors, or embeddings.
- Key challenge: new spam patterns appear over time (data drift), so evaluation and monitoring matter.

11 A minimal Pandas inspection snippet (illustrative)

```
import pandas as pd

df = pd.read_csv("data.csv")
print(df.shape) # (rows, columns)
print(df.head()) # first 5 rows
print(df.isna().sum()) # missing values per column
```

This basic habit catches many problems early.

12 Common pitfalls (and how to avoid them)

- **Treating categories as numbers:** encoding city as 1/2/3 creates a fake ordering; use one-hot (or other encodings).
- **Ignoring leakage:** features that contain future information make evaluation meaningless.
- **Confusing model and algorithm:** training procedure is the algorithm; the trained parameters are the model.
- **Assuming more features always helps:** irrelevant/noisy features can hurt generalization (dimensionality issues).

13 Practice (with answers)

P1. A dataset has 1,000 rows and 25 columns. If 1 column is the label, what is n and d?

Answer: $n = 1000$. If there are 25 total columns and 1 is the label, then $d = 24$ features.

P2. Give one example of a categorical feature and how you would encode it.

Answer: Feature: city={Delhi, Dehradun, Mumbai}. Encode using one-hot: three binary columns, one per city.

P3. Why can accuracy be misleading on imbalanced data?

Answer: If 95% examples are class 0, a model that always predicts 0 gets 95% accuracy but is useless for detecting class 1.

14 Exit questions (with answers)

Q1. Define ML in your own words.

Answer: ML is a set of methods that learn patterns from data to make predictions or decisions, rather than being explicitly programmed with rules.

Q2. Features vs labels?

Answer: Features are input variables (X). Labels/targets are outputs (y) that the model predicts.

Q3. Give two supervised and two unsupervised tasks.

Answer: Supervised: spam classification, house price regression. Unsupervised: customer clustering, PCA for dimensionality reduction.

Q4. Name three common dataset issues.

Answer: missing values, outliers, noisy/incorrect labels, duplicates, inconsistent formatting, class imbalance.

15 Summary

ML depends on datasets and good feature representations. The next lecture focuses on splitting and evaluation to avoid misleading results.

Sources: Mueller and Massaron, *Machine Learning for Dummies*, For Dummies, 2016.;Mitchell, *Machine Learning*, selected chapters on learning basics and evaluation.;Scikit-learn documentation, accessed 2026-02-28.